

CSE-422 Project Report

Predictive Modeling and Analysis for Depression Risk Using Machine Learning Techniques

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1 Introduction

1.1 Project Aim

This project aims to develop and evaluate predictive machine learning models capable of classifying the risk and severity of depression in individuals. By analyzing demographic details, academic metrics, and responses to psychometric questionnaires, the project seeks to build an automated diagnostic tool using both supervised algorithms and unsupervised clustering techniques.

1.2 Problem Statement

Depression is frequently underdiagnosed or detected late due to a heavy reliance on manual, subjective screening processes. Traditional diagnostic workflows are resource-intensive and lack the scalability required for broad, early-warning detection. The technical challenge is to identify distinct risk patterns and accurately classify individuals while minimizing false negatives in the dataset.

1.3 Motivation

Mental health disorders are a leading cause of global health burdens. The motivation behind this project is to leverage machine learning to discover complex relationships between academic pressure, behavioral habits, and mental state. An automated predictive model can serve as an early-intervention aid for counselors and healthcare providers, helping identify high-risk individuals before severe clinical escalation occurs.

2 Dataset Description

- **Total Features:** The cleaned dataset contains 40 features (after removing data leakage and applying One-Hot Encoding).
- **Problem Type:** This is a **classification** problem because the target variable (Depression Label) consists of distinct, discrete categories (severity levels from 0 to 5) rather than a continuous numerical scale.
- **Total Data Points:** 1,971 valid instances (after dropping 6 rows with missing values).
- **Feature Types:** The dataset contains both Quantitative (ordinal survey responses, aggregate scores) and Categorical (Demographics, Academic Year) features.
- **Encoding:** Categorical variables must be encoded because machine learning algorithms rely on mathematical equations and distance calculations to learn patterns. We used Ordinal Mapping for the target variable to preserve severity order, and One-Hot Encoding for the demographic inputs to prevent false mathematical hierarchies.

2.1 Feature Correlation



Figure 1: Correlation Heatmap of Features and Target Variable

Based on the correlation heatmap, several key insights emerge:

- **Grouped questions are strongly linked:** Dark red blocks show that questions on the same topic (like anxiety) are highly correlated. High scores on one usually mean high scores on the rest.
- **Stress, Anxiety, and Depression are connected:** Red areas linking the total scores show that students with high academic stress or anxiety also tend to have higher depression scores.
- **Confidence lowers depression (Negative Correlation):** Blue areas highlight an opposite relationship. As a student's confidence and sense of control increase, their stress and depression scores decrease.

3 Imbalanced Dataset & Exploratory Data Analysis

3.1 Imbalanced Dataset

For the output feature, the unique classes do not have an equal number of instances. The dataset is heavily skewed toward the higher severity categories, with **Moderately Severe Depression** (495 instances) and **Severe Depression** (488 instances) forming the vast majority, while **No Depression** contains just 44 instances.

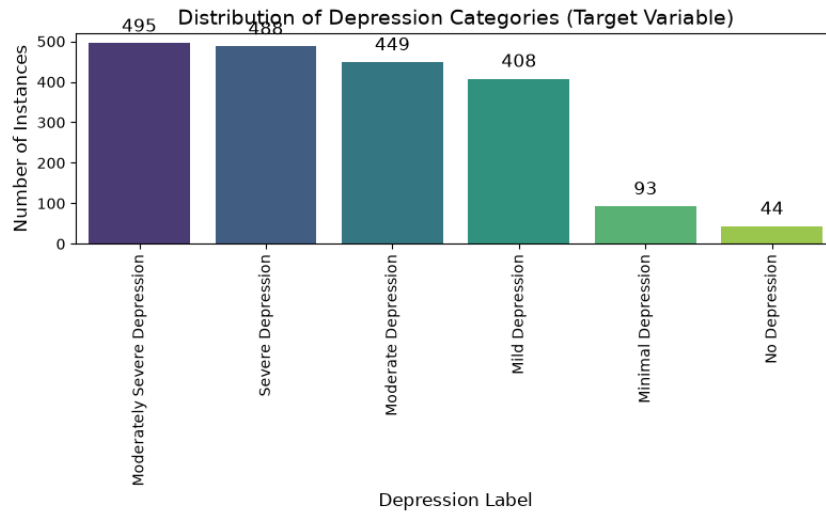


Figure 2: Distribution of Depression Categories Showcasing Class Imbalance

3.2 Exploratory Data Analysis (EDA)

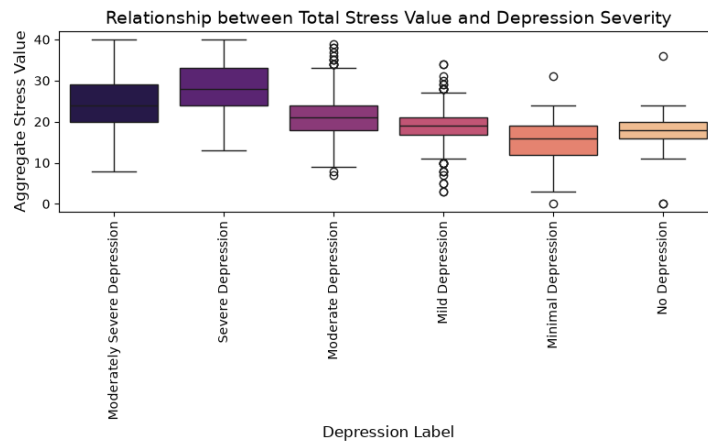


Figure 3: Relationship between Total Stress Value and Depression Severity

Depression Severity vs. Stress Levels: A boxplot analyzing the relationship between the **Aggregate Stress Value** and **Depression Label** shows a distinct upward trend, confirming that aggregate stress is a strong positive indicator of depression severity.

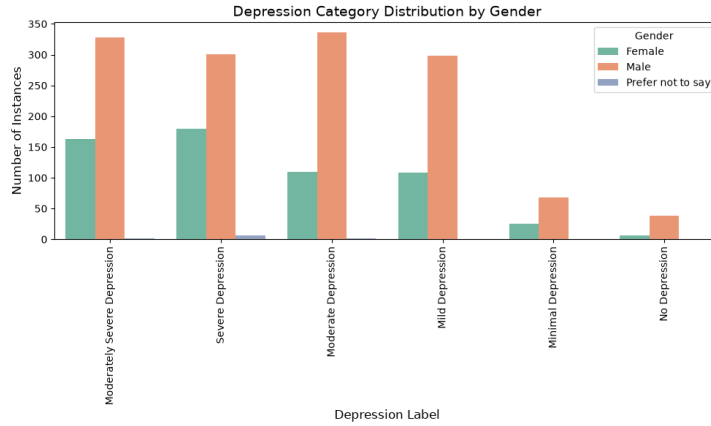


Figure 4: Depression Category Distribution by Gender

Depression Distribution by Gender: A grouped bar chart reveals a significant gender disparity. Across the severe depression categories, the number of male instances is significantly higher than female instances, highlighting a demographic skew in the sampled population.

4 Dataset Pre-processing

4.1 Problem 1: Null / Missing Values

Upon inspecting the dataset, the **Gender** column contained 6 missing values out of 1,977 total instances.

Solution Applied (Row Deletion): Because 6 rows represent a statistically negligible fraction (approx. 0.3%), imputing these values could introduce synthetic bias. The rows were completely deleted to ensure the model trains only on complete, authentic data.

4.2 Problem 2: Categorical Values

Machine learning algorithms require numerical input to compute mathematical weights. This dataset contains categorical features stored as strings.

Solution Applied (Ordinal & One-Hot Encoding): Ordinal Mapping was applied strictly to the target variable to create a 0-5 severity scale. For input features like **Gender**, **One-Hot Encoding** (`pd.get_dummies`) was applied, transforming them into separate binary columns to ensure the models treat all demographic groups equally.

4.3 Problem 3: Data Leakage

The aggregate **Depression Value** and the 9 raw depression questions mathematically added up to the exact target label, causing deterministic mapping (data leakage).

Solution Applied (Column Deletion): To force the algorithms to create a true predictive model based on external factors, these 10 leakage columns were explicitly dropped from the input features (X).

4.4 Problem 4: Feature Scaling

Numerical features have varying ranges. To prevent distance-based algorithms from disproportionately weighing features with larger scales, scaling is required.

Solution Applied (Min-Max Normalization): Min-Max Scaling was applied. Crucially, to prevent data leakage, this scaling was executed exclusively *after* the dataset was split.

5 Dataset Splitting

- **Split Ratio:** An 80/20 split was utilized (1,576 instances for the Training Set, 395 instances for the Testing Set).
- **Splitting Strategy (Stratified):** Because the dataset is imbalanced, Stratified Sampling was applied during the split. This guarantees that the exact class proportions of the original dataset are preserved across both the training and testing sets, ensuring fair model evaluation.

6 Model Training & Testing (Supervised & Unsupervised)

6.1 Supervised Learning

Three supervised algorithms were trained to predict the `Depression Label`:

1. **Logistic Regression:** Selected as the baseline linear model for this classification task.
2. **Decision Tree Classifier:** Selected for its non-linear, rule-based approach.
3. **Neural Network (Multi-Layer Perceptron):** A feedforward Artificial Neural Network with two hidden layers (64 and 32 neurons) was deployed to capture complex, non-linear relationships.

6.2 Unsupervised Learning (K-Means Clustering)

Treating the problem as an unsupervised learning task, the K-Means clustering algorithm was applied to the scaled feature set (excluding labels). The number of clusters (K) was explicitly set to 6. Principal Component Analysis (PCA) was utilized to map the 40 dimensions down to 2 principal components, allowing the 6 clusters to be visually showcased on a 2D scatter plot.

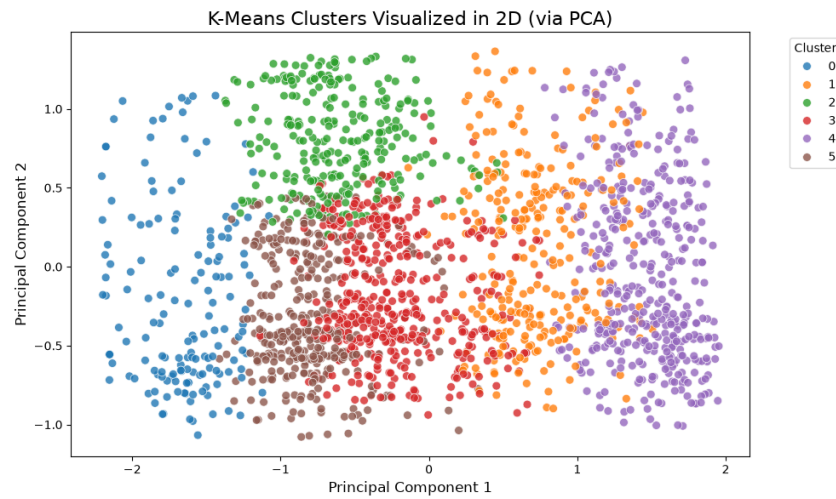


Figure 5: K-Means Clusters Visualized in 2D using PCA

7 Model Selection & Comparison Analysis

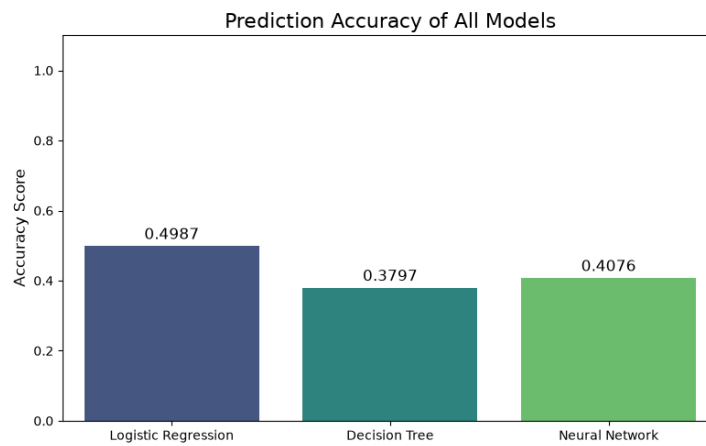


Figure 6: Prediction Accuracy of All Models

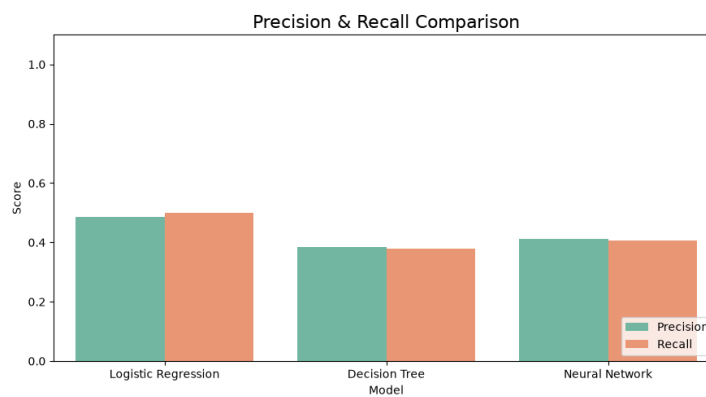


Figure 7: Precision & Recall Comparison

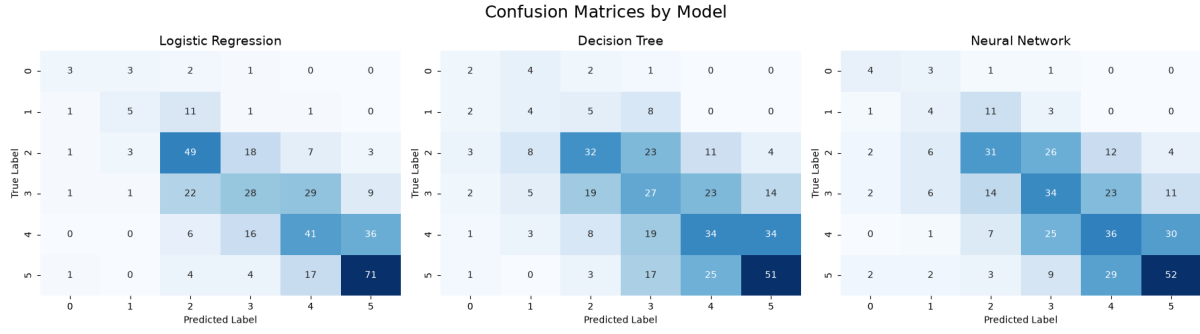


Figure 8: Confusion Matrices by Model

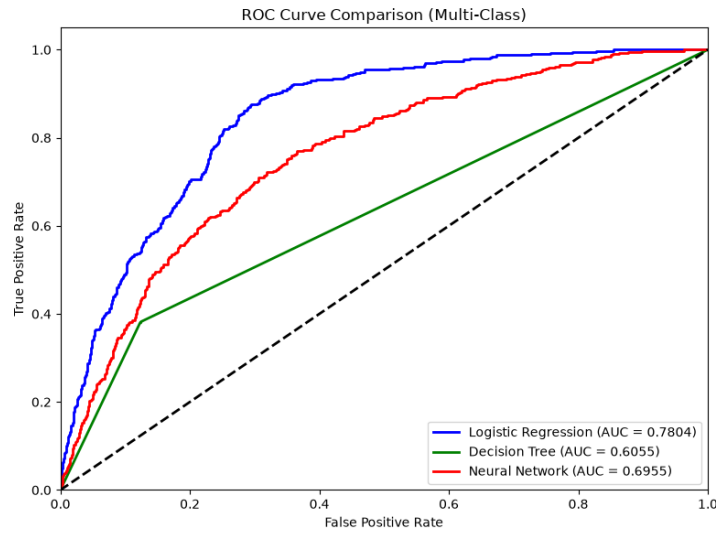


Figure 9: ROC Curve & AUC Scores (Multi-Class)

Based on the calculated metrics, removing the data leakage transformed this into a highly challenging, true predictive modeling task:

- Accuracy, Precision, & Recall:** Logistic Regression performed the best, achieving an accuracy of 0.4987 (Precision: 0.4852, Recall: 0.4987). This is exceptionally strong for a 6-class problem where random guessing yields approximately 16.6%. The Neural Network followed at 0.4076, while the Decision Tree overfit the training data and struggled on the testing set, scoring 0.3797.
- ROC Curve & AUC Score:** Logistic Regression led with the highest AUC score of 0.7805, indicating a strong ability to separate the different severity classes probabilistically. The Neural Network achieved a 0.6955 AUC, while the Decision Tree lagged at 0.6055.
- Confusion Matrix:** The confusion matrices reveal the difficulty of a 6-class psychological scale. However, Logistic Regression’s matrix shows a strong diagonal trend, indicating that even when it misclassified a student, it generally predicted an “adjacent” severity level (e.g., predicting “Severe” instead of “Moderately Severe”).

8 Conclusion

8.1 What we understand from the results

The results demonstrate that academic stress, anxiety, and demographic factors are strong, genuine predictors of depression risk. Even after removing all direct depression-related survey questions from the dataset, the baseline linear model could accurately predict a student’s exact depression severity tier nearly 50% of the time, proving the compounding effect of academic pressure on mental health. The unsupervised K-Means clustering also proved that individuals naturally group together based on these survey responses.

8.2 Comments regarding the performance of the models

Logistic Regression provided the best performance (49.87%). In highly subjective, psychometric datasets where features (like stress and anxiety) share strong, direct linear correlations with the target variable, simpler linear models often generalize better to unseen data than highly complex models like Neural Networks or prone-to-overfitting Decision Trees.

8.3 Reason for getting such results: "Fixing Data Leakage"

During early iterations, algorithms achieved a perfect 1.0000 (100%) accuracy. This highlighted a classic data science phenomenon known as **data leakage**. Because the original input dataset retained the aggregated **Depression Value**—which is the exact numerical sum used to assign the **Depression Label**—the algorithms simply memorized the grading rubric. To create a *true* predictive scenario, we explicitly dropped the **Depression Value** and the 9 raw depression questions. The current results (50% accuracy) represent a genuine, leakage-free prediction of depression risk based purely on a student’s stress and anxiety markers.

8.4 Challenges we faced

1. **Preventing Data Leakage:** Identifying and removing the specific features that deterministically mapped to the target variable to ensure the models were actually “learning” rather than just “memorizing.” We also had to ensure Min-Max scaling was applied strictly *after* the train/test split to prevent testing data from leaking into the training scaler.
2. **Imbalanced Dataset:** The dataset skewed heavily toward severe depression cases, with very few “No Depression” instances. This required implementing Stratified Splitting to ensure the testing phase was fair and unbiased.
3. **Repetitive Categorical Features:** The raw dataset contained long, repetitive sentence strings for column names and categorical text for labels, requiring extensive data cleaning, ordinal mapping, and One-Hot Encoding before the algorithms could process the mathematics securely.